

- (ii) Determine the value of θ at which the second-order bright fringe is formed.

$$\theta = \dots\dots\dots^\circ \quad [1]$$

- (iii) On Fig. 4.2, sketch the variation of the intensity I with θ for values of θ from -15° to $+15^\circ$.

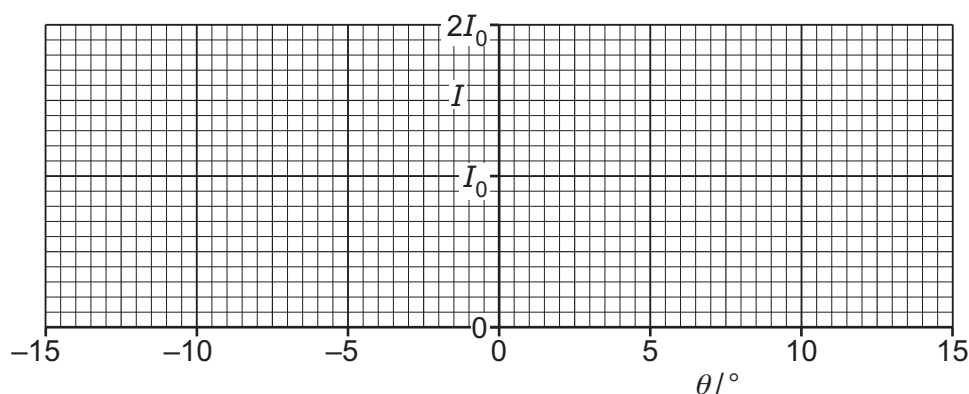


Fig. 4.2

[3]

- (c) A polarising filter is placed in the path of the light beam that is incident on the diffraction grating in Fig. 4.1. The transmission axis of the filter is at 45° to the vertical.

Suggest how the variation of intensity with θ for the light on the screen compares with the answer in (b)(iii).

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 [2]

[Total: 10]

